An Algebra of Prediction-Rewarding Mechanisms

Scoring rules, market makers, and pools as composable transducers

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Abstract

Scoring rules, market makers, parimutuels, and opinion pools compose when each stage is a stateful transducer over one message type, and the message deployed contests actually collect is a finite cloud of samples. This note provides sample-based elicitation inside that algebra: the companion paper's jittered settlement makes cloud submission strictly proper; here the composition operators act on clouds pointwise, and joint laws factor into margin stages plus a copula stage settled on the rank vector. The single-stage theory underneath is classical convex duality: proper scores from convex entropies, cost-function market makers by Fenchel conjugacy, the two opinion pools as the two Kullback-Leibler barycenters, merged market makers as the infimal convolution of risk sharing. The benefit of composing is illustrated by the simplest two-stage example: a residual market collects the conformal predictor's information gap $I(R; X)$ as bankroll growth while its marginal coverage stays exact.

Keywords: proper scoring rules; cost-function market makers; parimutuel mechanisms; opinion pools; convex duality; mechanism composition; conformal prediction markets

1 Introduction

Mechanisms for eliciting and aggregating forecasts are usually studied one at a time. A proper scoring rule is analysed as a one-shot contract; a market maker as a sequential trading venue; an opinion pool as an estimator; a calibration test as a diagnostic. Deployed systems are thinner than the theory allows: with few exceptions they run one level of competition against one internal model. Numerai pools staked submissions into a single stake-weighted meta-model and pays each forecast its marginal contribution to it (Craib et al. 2017); CrunchDAO blends each contest into one ensemble; the IARPA prediction polls aggregated forecasts by track record, comparing favourably with head-to-head markets (Atanasov et al. 2017), with reputation rather than wealth as the threaded state. In every case, one pool and one internal aggregate.

Adjacent markets contain pieces of the structure, none of them the piece this note needs. Derivatives chain point outputs: every futures contract settles on another market's price, and in electricity the virtual bids and transmission rights that settle on day-ahead prices pay forecasters for correcting the day-ahead consensus (Jha and Wolak 2023). These are translations of a price, fixed transforms of a point output, not composition of elicitation mechanisms. The racetrack tote prices margins and joint finishing orders in parallel books, neither settling on the other's output, consistency left to arbitrage (Harville 1973; Hausch et al. 1981). Reuse of

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a market's *probabilistic* output, a distribution, percentile, or rank emitted by one stage becoming the message or the settlement transform of the next, is scarce, and is the chain this note studies. Its one deployed instance ran on the microprediction platform (Cotton 2022): streams spawned z-streams of community percentiles, and bivariate and trivariate dependence streams priced copulas, so calibration and dependence were themselves the subject of further games; the stacked-lottery design behind it was presented at MIT CSAIL in 2020 (Cotton 2020, slides 29-31), and the platform is retired. The nearest live thing is monteprediction, a weekly self-funding pool over million-scenario joint submissions in eleven dimensions with wealth threaded across rounds since January 2024 (Cotton 2024b): the Sequentialise operator of §4 in production, one stage repeated, not a chain.

What a stage elicits moves under transformation of its message or its outcome. Scoring a kernel-smoothed submission at the raw outcome elicits the deconvolution of the belief rather than the belief; jittering the settlement by the smoothing kernel repairs it (Cotton 2026a). This is a property of a single stage, not of the chain, and it is the third question below. The repair makes a finite sample cloud a legitimate message type, and §6 runs the algebra on the objects deployed contests actually collect.

Conformal prediction supplies a second motivation, from the opposite direction. Split-conformal is itself a composition, a point predictor chained into a rank-based calibration stage, but a degenerate one: the pool step is skipped by fiat, all credit assigned to a single model in advance, and the calibration stage prices the residual flat in the input. The wasted log score of a single-shape conformal predictor, one that applies the same residual law at every input, is the mutual information $I(R;X)$ between residual and input, and an entrant to a parimutuel residual pool who conditions on the input collects it as bankroll growth at exactly that rate (Theorem 9). Running the residual stage as an actual pool, rather than assuming its winner, is what the operators below are for; Proposition 4 locates the gap in the theory of the probability integral transform, and §5 prices it.

This note organizes the catalogue around three questions that are often blurred together:

1. *Message closure.* Can every stage consume and emit one common object, so that stages plug together at all?
2. *Convex generation.* Is each individual mechanism generated by a convex potential, so that one dictionary covers scoring rules, market makers, and their duals?
3. *Propriety under transformation.* When a stage transforms forecasts or outcomes, does strict propriety survive?

The first is an engineering convention (§2, §8). The second and third admit theorems, which are stated with complete proofs in §3 and §4; that mathematics is classical, collected in one place and one convention. The contributions are the parts built on it: sample-based elicitation, running the algebra on clouds rather than exact densities (§6); the factoring of multivariate elicitation into margin stages and a rank-settled copula stage (§7); and the priced conformal example (§5), where the residual market's bankroll growth is the conformal predictor's information gap. Play is stagewise throughout: no participant deviates across stages (§4). Cross-stage strategy is outside the note's scope.

2 Preliminaries

The finite setting. Section 3 works with a finite outcome set $\{1, \dots, n\}$ and reports p in the probability simplex Δ; from the aggregation operators of §4 onward the paper moves to the continuous setting, densities and distribution functions on $\mathbb{R}$ or $[0,1]^d$. The finite statements transfer with the usual measure-theoretic care; the one place where the transfer is not routine is the probability integral transform, treated separately in Proposition 4.

Scores. A scoring rule assigns $S(p,i) \in \mathbb{R} \cup \{-\infty\}$ to a report p and outcome i; the expected score of reporting p under belief q is $S(p;q) = \sum_i q_i S(p,i)$, assumed well defined with $S(q;q)$ finite (this is the regularity used throughout). The rule is *proper* if $S(q;q) \geq S(p;q)$ for all p, q and *strictly proper* if equality forces $p = q$. Scores are written in reward form: higher is better.

Convex tools. For convex G with subgradient selection $G'(p) \in \partial G(p)$, the Bregman divergence is $D_G(q,p) = G(q) - G(p) - \langle G'(p), q - p \rangle$. The Legendre-Fenchel conjugate of a function R on Δ is $R^*(\mathbf{q}) = \sup_{p \in \Delta} \langle p, \mathbf{q} \rangle - R(p)$. The infimal convolution of f and g is $(f \,\square\, g)(x) = \inf_y f(y) + g(x-y)$. "Closed proper convex" is used in the standard sense (Rockafellar 1970). Differentiable statements are on the relative interior of Δ, with extended-real values allowed on the boundary; subgradients on Δ act on its tangent space and are defined up to addition of multiples of **1**.

3 One potential, three mechanisms

Theorem 1 (characterisation of proper scoring rules; Savage (1971), McCarthy (1956)). *A regular scoring rule S is proper iff there is a convex $G : \Delta \to \mathbb{R}$ with*

$$S(p,i) \;=\; G(p) + \langle\, G'(p),\; e_i - p \,\rangle, \qquad G'(p) \in \partial G(p),$$

where e_i is the i-th unit vector. It is strictly proper iff G is strictly convex relative to Δ, and then $G(p) = S(p;p)$ is the expected score of a truthful forecaster.

Proof. ($\Leftarrow$) Since $\sum_i q_i(e_i - p) = q - p$, the expected score is affine in the belief: $S(p;q) = G(p) + \langle G'(p), q - p \rangle$. Hence

$$S(q;q) - S(p;q) \;=\; G(q) - G(p) - \langle G'(p),\, q - p \rangle \;=\; D_G(q,p),$$

which is non-negative by the supporting-hyperplane inequality, and zero only at $q = p$ when G is strictly convex relative to Δ; so S is (strictly) proper.

($\Rightarrow$) Define $G(q) := S(q;q)$. Properness says $G(q) = \sup_p S(p;q)$, and each $q \mapsto S(p;q) = \langle S(p,\cdot), q \rangle$ is affine, so G is a pointwise supremum of affine functions, hence convex. For any fixed p,

$$S(p;q) \;=\; G(p) + \langle S(p,\cdot),\, q - p \rangle \quad \text{with} \quad S(p;q) \leq G(q) \;\; \forall q,$$

so the vector $S(p,\cdot)$ is a subgradient of G at p (acting on the tangent space of Δ, where subgradients are defined up to a constant shift along **1**, which the representation absorbs). Substituting $G'(p) = S(p,\cdot)$ into the display returns $S(p,i)$ identically. If S is strictly proper the supremum has a unique maximiser at every q, which for a supremum of affine functions is equivalent to strict convexity relative to Δ. ■

The content of Theorem 1 is the dictionary *proper scoring rule* $\leftrightarrow$ *convex function* $\leftrightarrow$ *Bregman divergence* (Gneiting and Raftery 2007; Banerjee et al. 2005). The classics are three choices of G, with the divergences computed directly from the definition:

Score	Generator $G(p) = S(p; p)$	Bregman divergence $D_G(q, p)$
logarithmic	$\sum_i p_i \log p_i$	$\mathrm{KL}(q \\| p)$
Brier (quadratic)	$\\|p\\|_2^2$	$\\|q - p\\|_2^2$
spherical	$\\|p\\|_2$	$\\|q\\|_2 \, (1 - \cos\theta_{p,q})$

For the spherical row: $\nabla G(p) = p/\|p\|$, so $D_G(q, p) = \|q\| - \langle p, q\rangle/\|p\| = \|q\|(1 - \cos\theta_{p,q})$, an angular term scaled by $\|q\|$. For the logarithmic row the representation gives $S(p, i) = \log p_i$ on the relative interior, with $-\infty$ on the boundary.

Theorem 2 (scoring rule to market maker; Hanson (2007), Abernethy et al. (2013)). *Let R be closed, proper, and strictly convex on Δ, extended by $+\infty$ off Δ so that $C = R^*$ below is the Fenchel conjugate on $\mathbb{R}^n$ (in the sequel, $R = G$, the generator of Theorem 1 read as a regulariser), and define*

$$C(\mathbf{q}) = \sup_{p \in \Delta} \left(\langle p, \mathbf{q} \rangle - R(p) \right).$$

Then:

(i) C is convex and finite, the maximiser $p^*(\mathbf{q})$ is unique, and $\nabla C(\mathbf{q}) = p^*(\mathbf{q}) \in \Delta$: prices are a probability vector. (Without strict convexity, prices live in $\partial C(\mathbf{q})$.)

(ii) C is translation-equivariant, $C(\mathbf{q} + \alpha\mathbf{1}) = C(\mathbf{q}) + \alpha$; costs telescope over any trade path, so round trips cost zero and buying the full bundle costs its payout: the market is arbitrage-free.

(iii) With initial state $\mathbf{q}_0 = \mathbf{0}$, the maker's loss when the market settles on outcome i after net sales $\mathbf{q}$ is

$$\mathrm{loss}_i(\mathbf{q}) = q_i - C(\mathbf{q}) + C(\mathbf{0}) \ \le \ R(e_i) - \inf_{p \in \Delta} R(p) \ \le \ \sup_{p \in \Delta} R(p) - \inf_{p \in \Delta} R(p).$$

(iv) Taking $R(p) = b \sum_i p_i \log p_i$ gives $C(\mathbf{q}) = b \log \sum_i e^{q_i/b}$, Hanson's LMSR, with worst-case loss $b \log n$.

Proof. (i) C is a supremum of affine functions of $\mathbf{q}$, hence convex; the supremum of a continuous function over the compact Δ is attained, and strict concavity of $p \mapsto \langle p, \mathbf{q}\rangle - R(p)$ makes the maximiser unique. Danskin's theorem gives $\nabla C(\mathbf{q}) = p^*(\mathbf{q})$. (ii) $\langle p, \mathbf{q} + \alpha\mathbf{1}\rangle = \langle p, \mathbf{q}\rangle + \alpha$ for $p \in \Delta$, so the supremum shifts by α. A trader moving the state $\mathbf{q} \to \mathbf{q}'$ pays $C(\mathbf{q}') - C(\mathbf{q})$ by definition, so costs over any path telescope; a round trip costs zero, and by translation equivariance the bundle $\alpha\mathbf{1}$ costs exactly α, its sure payout. (iii) The maker collects $C(\mathbf{q}) - C(\mathbf{0})$ and pays q_i, so $\mathrm{loss}_i = q_i - C(\mathbf{q}) + C(\mathbf{0})$. By Fenchel-Moreau, $\sup_{\mathbf{q}} \left(\langle e_i, \mathbf{q}\rangle - C(\mathbf{q}) \right) = R^{**}(e_i) = R(e_i)$, and $C(\mathbf{0}) = \sup_p -R(p) = -\inf_p R(p)$: by Fenchel-Moreau the first bound is the exact supremum of the loss over $\mathbf{q}$. The final inequality holds because $e_i \in \Delta$. (iv) Lagrange: maximising $\langle p, \mathbf{q}\rangle - b \sum p_i \log p_i$ subject to $\sum p_i = 1$ gives $q_i - b(\log p_i + 1) = \lambda$, so $p_i \propto e^{q_i/b}$, and substituting back yields $C(\mathbf{q}) = b \log \sum_i e^{q_i/b}$. Then $R(e_i) = 0$ and $\inf R = -b \log n$ at the uniform distribution, so the loss bound is $b \log n$. ■

Proposition 3 (cost-function markets and CFMMs). *Under the monotonicity, concavity, and reserve-domain hypotheses of Frongillo et al. (2024), cost-function prediction markets and constant-function market makers can be converted into one another. In the unconstrained-reserve sign convention a cost function C gives a concave CFMM potential by $\varphi(\mathbf{r}) = -C(-\mathbf{r})$; bounded-reserve versions require a perspective (level-set) construction. The equivalence is a convex level-set duality; it is not the bare Fenchel conjugacy $C \mapsto C^*$.*

We do not reprove the general equivalence here; Frongillo et al. (2024) give it in full, and Angeris et al. (2023) and Angeris et al. (2021) develop the CFMM side.

Example (constant product). Take two outcome tokens with reserves r_1, r_2 and invariant $r_1 r_2 = k$. The pool's portfolio value at prices $(p, 1-p)$ is

$$V(p) = \inf\{p\, r_1 + (1-p)\, r_2 : \; r_1 r_2 = k\} = 2\sqrt{k\, p(1-p)},$$

by the AM-GM inequality, with the infimum attained at $r_1 = \sqrt{k(1-p)/p}$, $r_2 = \sqrt{kp/(1-p)}$. V is concave; reading $R(p) = -V(p)$ as the regulariser of Theorem 2 gives a maker whose worst-case loss is the range of R on $[0,1]$, namely $\sqrt{k}$ (between $p = \frac{1}{2}$ and the endpoints): the geometric mean of the initial reserves. The corresponding generator is not entropic, which is the convex-analytic content of the observation that constant-product markets and the LMSR price bounded-payout claims differently.

Proposition 4 (the probability integral transform; Rosenblatt (1952), Dawid (1984)). *If X has continuous CDF F then $U = F(X) \sim \mathrm{Uniform}(0,1)$, and $z = \Phi^{-1}(U) \sim N(0,1)$. In the prequential setting, if F_t is the true conditional law of X_t given the past, then $U_t = F_t(X_t)$ is conditionally uniform and the PIT stream is iid uniform.*

Proof. For continuous F, with the generalized inverse $F^{-}(u) = \inf\{x : F(x) \geq u\}$, the event $\{F(X) \leq u\}$ differs from $\{X \leq F^{-}(u)\}$ by a set of probability zero, and $\Pr(X \leq F^{-}(u)) = F(F^{-}(u)) = u$ by continuity. The prequential statement applies this conditionally at each t. ■

Two scope remarks. First, the converse fails in the strong sense: marginally uniform PITs do not certify an informative forecast. Reporting the unconditional law F of an iid sequence gives $F(X_t) \sim \mathrm{Uniform}(0,1)$ even if valuable covariates were ignored. A PIT critic therefore witnesses miscalibration relative to its test class; it complements, and does not replace, a proper score that rewards sharp conditional distributions. Conformal prediction lives on exactly this distinction: a split-conformal predictor achieves marginal coverage by construction, the uniform-PIT guarantee, while free to ignore the conditional information in the input; §5 prices the gap (Theorem 9). Second, for discrete forecasts the randomized PIT preserves the exact uniform null, while the mid-PIT is a convenient deterministic diagnostic with a different null distribution.

4 Operators on mechanisms

The common signature. A *transducer* (Mealy machine) is a tuple $(S, A, B, \delta, \lambda)$ with state space S, input alphabet A, output alphabet B, transition map $\delta : S \times A \to S$, and output map $\lambda : S \times A \to B$; run on an input stream it produces $s_{t+1} = \delta(s_t, a_t)$ and $b_t = \lambda(s_t, a_t)$, a causal map of input streams to output streams. The mechanisms of §3 share one instantiation. Let Dist denote the set of distributional beliefs and take S the wealth states, $A = \mathrm{Dist}^m \times \mathcal{X}$

(m participant reports and, where the stage settles, a realized outcome), and $B = \mathrm{Dist} \times \mathbb{R}^m$ (an aggregate belief and transfers). A *stage* is such a transducer, written

$$M : (\mathrm{Dist}^m,\ w,\ x) \longmapsto (\mathrm{Dist},\ w',\ \pi).$$

A scoring rule is the transfer component λ of a stage, not itself a map Dist $\to$ Dist; a market maker is a stage whose state is the inventory vector and whose emitted belief is the price; an opinion pool is a stage with no outcome argument and zero transfers. Composition wires the belief output of one stage to the belief inputs of the next while state and transfers thread through.

The operators below act on stages.

Sequentialise. Theorem 2: a proper score run sequentially against a wealth state is a cost-function market maker.

Pool. A proper score gives a batch elicitation mechanism when reports are scored independently and funded externally: properness is inherited report by report. Parimutuel and budget-balanced versions are a different game, because the pot split couples payoffs through the denominator; the price-taking analysis (Cotton 2026a, sec. 2) gives truthful all-in, a symmetric equilibrium at fractional stakes, and degeneracy as the stake fraction vanishes, and beyond that the equilibrium theory is open.

Ensemble (Proposition 5: the two pools are the two KL barycenters). *Let $q_1, \dots, q_m$ be densities with respect to a common dominating measure, $w_i \geq 0$, $\sum w_i = 1$, and for the second display assume $0 < \int \prod_i q_i^{w_i} < \infty$. Then the linear pool minimizes the forward divergences and the logarithmic pool the reverse:*

$$\arg\min_p \textstyle\sum_i w_i\, \mathrm{KL}(q_i \| p) = \sum_i w_i q_i, \qquad \arg\min_p \sum_i w_i\, \mathrm{KL}(p \| q_i) \;\propto\; \prod_i q_i^{w_i}.$$

Proof. For the first, $\sum_i w_i\, \mathrm{KL}(q_i \| p) = \mathrm{const} - \int (\sum_i w_i q_i) \log p$, and $\int \bar{q} \log p$ is maximized over densities at $p = \bar{q}$ (Gibbs). For the second, $\sum_i w_i\, \mathrm{KL}(p \| q_i) = \int p \log p - \int p\, \overline{\log q}$ with $\overline{\log q} = \sum_i w_i \log q_i$; the Lagrange condition is $\log p = \overline{\log q} + \mathrm{const}$. ■

Training weights by cumulative log score and then mixing linearly, $\sum_m w_m F_m$, is Bayesian model averaging, a linear pool with score-trained weights; the logarithmic pool multiplies densities and renormalizes. They are different aggregates with different sharpness (Genest and Zidek 1986).

Merge (Proposition 6: merging makers is infimal convolution; Rockafellar (1970), Bhaskara et al. (2023)). *For closed proper convex f, g: $(f \,\square\, g)^* = f^* + g^*$. Consequently, merging two cost-function makers with regularisers R_1, R_2 (cost functions $C_i = R_i^*$) yields the maker with regulariser $R_1 + R_2$, and merging* LMSR_{b_1} *with* LMSR_{b_2} *yields* $\mathrm{LMSR}_{b_1+b_2}$*: liquidity adds.*

Proof. $(f \,\square\, g)^*(p) = \sup_x \langle p, x \rangle - \inf_y \{f(y) + g(x-y)\} = \sup_{y,z} \langle p, y \rangle - f(y) + \langle p, z \rangle - g(z) = f^*(p) + g^*(p)$. For the makers, $C_1 \,\square\, C_2 = (R_1 + R_2)^*$ by biconjugacy, and $b_1 G + b_2 G = (b_1 + b_2) G$ for the entropic generator. ■

This is also the risk-sharing literature's composition law: the aggregate of several agents' convex risk measures is their infimal convolution, with the Pareto allocation as minimiser and the common subgradient as the clearing price (Barrieu and El Karoui 2005; Jouini et al. 2008); the market-liquidity reading is developed by Bhaskara et al. (2023).

Conjugate. Run a stage in a transformed coordinate and map back. This is where propriety needs care, and the theory splits in two:

- *Fixed transformations.* Properness is preserved under any fixed transformation of forecast and outcome, and strict propriety iff the transformation is injective (Allen et al. 2023, Prop. 4; Pic et al. 2025, Prop. 1). The Markov-kernel (stochastic channel) extension, strict propriety iff the channel is injective on laws, is Theorem 2 of the companion point-cloud paper, whose Theorem 1 also exhibits the canonical failure: a KDE smoothing seam whose raw-outcome score elicits a deconvolution.
- *Forecast-dependent transformations.* The PIT critic transforms outcomes by the reported F itself, so the fixed-transformation theorems do not apply to it; Proposition 4 and its scope remarks are the correct warrant, and the critic is a calibration diagnostic rather than a strictly proper score. The stacked-lottery design (Cotton 2020, slides 29-31) is this operator in practice: percentiles from one game feed the next, and calibration is produced by composing monotone maps contributed by competing algorithms.

Residual. Let a stage emit the aggregate F_1 for an outcome Y, and let a second market elicit a distribution for the residual $U = F_1(Y)$, settling at the realised $u = F_1(y)$. If F_1 is the true conditional law then U is uniform (Proposition 4) and the second market has nothing to price; whatever structure remains in the residual is the second stage's edge. The corrected forecast composes the two reports.

Proposition 7 (the correction is a multiplicative reweighting). *Let F_1 be strictly increasing onto $(0, 1)$ with density $p_1 > 0$ and let the residual market's consensus be a distribution H on $[0, 1]$ with $H(0) = 0$, $H(1) = 1$ and density g. The composed forecast $F = H \circ F_1$ has density*

$$p(y) \;=\; p_1(y)\, g(F_1(y)),$$

so $\log p(y) = \log p_1(y) + \log g(u)$ with $u = F_1(y)$: the chain's log score is the sum of stage log scores, and the residual stage is paid by a proper score on u alone.

Proof. Chain rule: $F'(y) = g(F_1(y))\, p_1(y)$; take logarithms. The residual score $\log g(u)$ is the logarithmic score of Theorem 1 applied to the report g and outcome u, hence strictly proper for the law of U. ■

Multiplying the density by a ratio fitted to what the current model gets wrong is the functional-gradient step of boosting under log loss (Mason et al. 1999; Friedman 2001), so a chain of residual markets is stagewise boosting with wealth as the learning rate. What Proposition 7 does not settle is the game across stages: who funds the residual pot, and whether a forecaster free to enter both stages prefers to withhold information from the first and sell it to the second (§9).

Spec. Serialise a pipeline to data and search over it; the mechanism analogue is a market over pipelines. Also open.

Stagewise play. Call a profile of reports a *stagewise equilibrium* if no participant gains by a deviation confined to a single stage, all other stages' reports held fixed *and the inputs and settlement transforms of every other stage clamped at their pre-deviation values*. This is the pipeline version of the myopic-trader assumption standard in the market-scoring-rule literature (Hanson 2007; Chen and Vaughan 2010). The clamp is not cosmetic: a stage's output is wired into later settlement transforms (the residual point $u = F_1(y)$, the rank vector of §7), so an upstream deviation moves downstream payoffs even when every downstream report is held fixed.

Proposition 8 (single-stage guarantees compose under clamped stagewise play). *Suppose each stage of a pipeline, taken in isolation with its inputs and settlement transform fixed, makes the truthful report a best response: strict propriety for externally funded scoring stages (Theorem 1), the* *sequential scoring of Theorem 2 for market stages, the price-taking pot-split analysis of the companion paper for parimutuel stages, and the residual score of Proposition 7 for correction stages. Then truthful reporting at every stage is a stagewise equilibrium of the pipeline. If moreover no participant reports to a stage upstream of one in which they hold a position (disjoint stage membership suffices), the clamp is vacuous for every feasible deviation, and truthful reporting survives unrestricted single-stage deviations.*

Proof. With the other stages' inputs and transforms clamped, a deviation confined to stage k changes the deviator's payoff only through stage k's transfer map, and the stage-k hypothesis makes the truthful report a best response. For the second claim: a stage-k deviation moves stage k's transfer and, through stage k's output, transfers strictly downstream of k; a deviator with no downstream position collects none of the latter, so the propagation is payoff-irrelevant to them. ■

Proposition 8 is not a Nash equilibrium of the composed game. When the same participant reports upstream and holds exposure downstream, the deviation propagates through the settlement transform, and a downstream stake is a derivative written on an upstream settlement, with the attendant incentives to distort the underlying (Kumar and Seppi 1992; Jarrow 1994; Hanson and Oprea 2009; Ostrovsky 2012). Disjoint membership, zero downstream exposure for upstream reporters, or exogenous freezing of the settlement transforms restore the proposition; the dynamic game without them is outside the note's scope.

5 Betting against a conformal predictor

The introduction claimed that split-conformal prediction is a degenerate composition and that running the residual stage as a market collects what the degeneracy wastes. This section proves it. The analysis is population-level throughout: laws are continuous, the ranking uses the true marginal CDF, and the finite-sample, exchangeability-based conformal construction enters only through the measurement caveat at the end of the section. A point predictor leaves a residual R; conformalization re-levels its marginal law. Write $U = F_R(R)$ for the PIT of the residual, F_R the marginal CDF of R, so U is marginally uniform whatever the model (Proposition 4), and let $g(u \mid x)$ be the conditional density of U given $X = x$. Since $R \mapsto F_R(R)$ is almost surely invertible, $I(U; X) = I(R; X)$, and because the marginal of U is uniform,

$$I(R; X) \;=\; \mathbb{E}_X \, \mathrm{KL}(P_{R|X} \,\|\, P_R) \;=\; \mathbb{E}_X \int_0^1 g(u \mid X) \log g(u \mid X) \, \mathrm{d}u.$$

The residual stage is the parimutuel of the companion paper run on the rank scale. Two facts about that pool are needed.

Lemma 1 (pool payoff). *Let the crowd's aggregate stake have density q on the outcome space, normalised to one, and let a price-taking entrant stake $\epsilon\, b(u)\, \mathrm{d}u$ alongside it. As $\epsilon \to 0$ the entrant's gross payoff per unit of their own wealth, after outcome u, is $W(u) = b(u)/q(u)$.*

Proof. Bin the outcome axis at width δ. The crowd stakes $q(u)\delta$ on the bin at u, the entrant $\epsilon\, b(u)\delta$. If the outcome lands there the pool is split in proportion to stake, so the per-unit payoff is $1/(q(u)\delta)$ up to $O(\epsilon)$, and the entrant collects $\epsilon\, b(u)\delta \cdot 1/(q(u)\delta) = \epsilon\, b(u)/q(u)$: per unit of entrant wealth, $b(u)/q(u)$. The bin width cancels and the limit is the Radon-Nikodym derivative $\mathrm{d}b/\mathrm{d}q$. ■

The cancellation is what makes a lottery on a point outcome well posed: the pool divides two vanishing quantities and the ratio survives, with no reference measure and no posted odds.

Lemma 2 (log-optimal growth). *If the outcome is drawn from p, the expected log-growth of an entrant staking b against a crowd staking q is*

$$\mathbb{E}_{U\sim p}\Big[\log\frac{b(U)}{q(U)}\Big] = \mathrm{KL}(p\|q) - \mathrm{KL}(p\|b) \;\le\; \mathrm{KL}(p\|q),$$

with equality iff $b = p$: the log-optimal stake is the truth.

Proof. Add and subtract $\log p$ inside the expectation; Gibbs' inequality kills the second term exactly at $b = p$. ■

The lemmas are the parimutuel form of the Kelly-Breiman log-optimality principle (Kelly 1956; Cover and Thomas 2006; Barron and Cover 1988); when the crowd prices the marginal, the growth of belief b is $I(R;X) - \mathbb{E}_X\,\mathrm{KL}(g(\cdot \mid X)\|b(\cdot \mid X))$, a decomposition written by Kemp and Bettencourt (2022) for population growth in stochastic environments.

Theorem 9 (parimutuel rent of a conformal predictor). *Put the pool on the rank scale. The single-shape conformal predictor is the crowd that prices it flat, $q \equiv 1$ on $[0,1]$. Against it:*

(i) an entrant who knows only the marginal stakes $b \equiv 1$ and grows at rate zero: marginal coverage stated as wealth;

(ii) an entrant who observes X and stakes $b = g(\cdot \mid X)$ grows per round at rate $\mathbb{E}_X\,\mathrm{KL}(g(\cdot \mid X)\|1) = I(R;X)$;

(iii) with a track take $\tau \in [0,1)$ the informed rate is $I(R;X) + \log(1-\tau)$, positive iff $I(R;X) > -\log(1-\tau)$.

Proof. Lemma 1 with $q \equiv 1$ makes the payoff $b(U)$; Lemma 2 makes the growth $\mathrm{KL}(p\|1) - \mathrm{KL}(p\|b)$ for true conditional law $p = g(\cdot \mid x)$. For (i), $b = 1$ and the marginal of U is uniform, so the rate is zero. For (ii), conditioning on $X = x$ the optimal stake is $g(\cdot \mid x)$ with conditional rate $\mathrm{KL}(g(\cdot \mid x)\|1) = \int g \log g$; averaging over X gives $I(R;X)$ by the display above. For (iii) every payoff is multiplied by $1-\tau$, adding $\log(1-\tau)$ to the rate. ■

The certificate and the leak are two readings of one pool. Marginal coverage is part (i), a break-even statement; part (ii) is what the certificate cannot see, and re-leveling cannot close it because re-leveling acts on the marginal, where the leak is invisible. Conformal re-leveling takes no rake, so any conditional information at all is pure profit.

Example (Gaussian residual). Let (R,X) be jointly Gaussian with correlation ρ, marginally $R \sim N(0,\sigma_R^2)$, $X \sim N(0,\sigma_X^2)$: the canonical case, left by a linear predictor whose residual retains correlation with the input. The informed entrant's report is $R \mid X = x \sim N(\rho(\sigma_R/\sigma_X)x,\,(1-\rho^2)\sigma_R^2)$, and with $m(x) = \rho(\sigma_R/\sigma_X)x$, $s^2 = (1-\rho^2)\sigma_R^2$,

$$\mathbb{E}_X\,\mathrm{KL}(N(m,s^2)\,\|\,N(0,\sigma_R^2)) = \mathbb{E}_X\Big[\log\frac{\sigma_R}{s} + \frac{s^2+m^2}{2\sigma_R^2} - \frac{1}{2}\Big] = -\tfrac{1}{2}\log(1-\rho^2) = I(R;X).$$

At $\rho = \frac{1}{2}$ the rent is 0.144 nats per observation and the informed bankroll doubles every five observations; at $\rho = 0.3$, every fifteen. Throughout, the conformal band's marginal coverage remains exact. The composed forecast of Proposition 7 built from the entrant's rank report $g(u \mid x) = p_{R|X}(F_R^-(u))/p_R(F_R^-(u))$ recovers the conditional law exactly: $p_R(r)\,g(F_R(r) \mid x) = p_{R|X}(r)$.

The mechanism has run in production. The microprediction platform's nearest-the-pin pool (Cotton 2022) paid each entry its sample density at the realised value relative to the field's, the payoff of Lemma 1 with densities estimated from submitted samples; the MidOne contest (CrunchDAO 2024) priced an explicit density (Cotton 2024a) the same way, and its priced object was a residual stream. A conformal predictor entering such a pool is the participant that prices flat in X, and Theorem 9 is the bankroll of its better-informed competitor.

Measuring the rent. The rent is defined through a log score, so it is estimated rather than read off. Two routes:

Proposition 10 (certified lower bound). *Let* $\mathrm{HSIC}(U, X)$ *be computed with a product kernel satisfying* $\sup_z k(z, z) \le K$. *Then* $I(R; X) \ \ge\ \mathrm{HSIC}(U, X)/(2K)$.

Proof. Write $\mathrm{TV} = \sup_A |P(A) - Q(A)|$ for the total variation between the joint law and the product of marginals. $\mathrm{MMD} = \mathrm{HSIC}^{1/2}$ is the integral probability metric over the unit ball of the tensor RKHS, whose witnesses satisfy $\|f\|_\infty \le \sqrt{K}$, so $\mathrm{MMD} \le 2\sqrt{K}\,\mathrm{TV}$. Pinsker's inequality in the same convention (Cover and Thomas 2006) gives $I(U; X) = \mathrm{KL} \ge 2\,\mathrm{TV}^2$, and $I(U; X) = I(R; X)$. Combining, $I(R; X) \ge 2(\mathrm{MMD}/2\sqrt{K})^2 = \mathrm{HSIC}/(2K)$. ■

Distance covariance (Székely et al. 2007) is a fixed multiple of an HSIC (Sejdinovic et al. 2013), so a measured dependence between rank and input certifies a minimum extractable rent. No matching upper bound exists with a fixed kernel: MMD metrizes weak convergence (Simon-Gabriel et al. 2023) and KL does not.

Proposition 11 (anytime-valid test and consistent estimate). *At round t choose a betting density* $b_t(\cdot \mid x) \ge 0$ *with* $\int_0^1 b_t(u \mid x)\,\mathrm{d}u = 1$ *from the history, and set* $W_t = \prod_{s \le t} b_s(U_s \mid X_s)$. *(i) Under the null of conditional rank uniformity,* $U_t \mid X_t, \mathcal{F}_{t-1} \sim \mathrm{Uniform}(0, 1)$, *the process* (W_t) *is a non-negative martingale with mean one, so* $\Pr(\sup_t W_t \ge 1/\alpha) \le \alpha$ *(Ville 1939): rejecting when* $W_t \ge 1/\alpha$ *is an anytime-valid level-α test of conditional rank uniformity, that is, of conditional dependence of the residual rank on the input. (ii) If the rounds are iid and* $\log b_t \to \log g$ *uniformly with a bounded envelope, then* $t^{-1} \log W_t \to I(R; X)$ *almost surely.*

Proof. (i) Under the null, each factor has conditional mean $\int_0^1 b_t = 1$; Ville's inequality applies to the resulting martingale. (ii) Write $\log b_s = \log g + (\log b_s - \log g)$: the second terms vanish in Cesàro mean by uniform convergence, and the average of the first tends almost surely to $\mathbb{E}[\log g(U \mid X)] = I(R; X)$ by the law of large numbers and the display opening this section. ■

The wealth process is simultaneously the test and the estimator; it is the log-optimal e-variable of safe testing (Grünwald et al. 2024; Vovk and Wang 2021) for the null $U \perp X$, with conformal test martingales as ancestor (Vovk et al. 2003). Sequential tests of forecast calibration (Arnold et al. 2023) and conditional independence by betting (Shaer et al. 2023) use the same mechanics; the identification of the growth rate with the conformal predictor's information gap, and with the realised bankroll in a deployed pool, is the reading here. One measurement caveat: $U = F_R(R)$ uses the true marginal, and with empirical ranks the null is exact in the full-conformal, leave-one-out sense. And one scope remark: by the distribution-free no-go results (Lei and Wasserman 2014; Foygel Barber et al. 2021; Vovk 2012) no procedure certifies conditional coverage from data alone, so a quiet e-process means no rent found at this power, never conditional validity.

6 Samples as messages

The message type has so far been a distribution given exactly. Deployed contests collect something rougher: a finite cloud of samples, smoothed by the contest into a density before settlement. The companion paper gives the sample-based elicitation result the algebra needs (Cotton

2026a):

Proposition 12 (sample-based elicitation; Cotton (2026a), Thms 1-2). *Score the bandwidth-h kernel density estimate of a submitted cloud by the logarithmic score. Settled at the raw outcome, the optimal cloud is drawn from a deconvolution of the belief when one exists (for Gaussian beliefs and kernels with belief variance exceeding h^2, the belief with h^2 removed from the variance). Settled at the outcome jittered by the same kernel, truthful sampling is optimal, and strictly so whenever the kernel's characteristic function is nonvanishing on a dense set: the smoothing channel is injective on laws.*

The proof is in the companion paper. With jittered settlement the cloud *is* a valid message, and the operators of §4 act on clouds directly.

- *Conjugate* acts pointwise: the pushforward of a cloud under ψ is ψ applied to each sample, with no Jacobian computed anywhere. This is the practical reason to prefer sample messages: transformation of densities requires calculus, transformation of clouds requires arithmetic.
- *Pool* (linear) is a wealth-weighted union: sample from participant i's cloud with probability proportional to w_i. The linear pool is the sample-native aggregate; the logarithmic pool has no comparably clean sample form, since multiplying densities requires importance weights.
- *Residual* acts by ranking: pass each sample through the upstream CDF. Composition of monotone maps, the stacked-lottery operation, is again pointwise on samples.
- *Sequentialise* is the nearest-the-pin pool itself: the market's state is the field's aggregate cloud.

One caveat governs the order of operations. Smoothing and conjugation commute only for affine maps: for scalar $\psi(x) = ax + b$ and a Gaussian kernel, the KDE of the mapped cloud at bandwidth $|a|h$ equals the pushforward of the bandwidth-h KDE (in higher dimension the bandwidth matrix must transform with the map), and for nonlinear ψ no bandwidth makes this an identity. The smoothing seam and its jitter must therefore be applied in the settlement coordinates, after all transforms, which in pipeline terms says the KDE stage belongs immediately before settlement and nowhere else.

7 Margins and copulas

A multivariate settlement can be factored. By Sklar's theorem (Sklar 1959) a joint law with continuous margins $F_1, \ldots, F_d$ and copula C has density

$$p(x) \;=\; \prod_{i=1}^{d} f_i(x_i) \;\cdot\; c(F_1(x_1), \ldots, F_d(x_d)),$$

with c the copula density on $[0,1]^d$.

Proposition 13 (the log score factors). *For a joint report assembled from margin reports f_i and a rank-stage report c, a density on $[0,1]^d$,*

$$\log p(x) \;=\; \sum_{i=1}^{d} \log f_i(x_i) \;+\; \log c(u), \qquad u_i = F_i(x_i),$$

so a pipeline of d margin stages and one rank stage settled on the rank vector u pays every stage a logarithmic score of its own object, and the chain's log score is the sum. Given the margin reports, the rank stage's score is strictly proper for the law of the rank vector. That law has uniform margins, and is the copula of the joint, iff the margin reports are correct; keeping the rank stage's report class as arbitrary densities on $[0,1]^d$ keeps the stage closed under wrong upstream reports.

Proof. Take logarithms in Sklar's density factorization; the summands are exactly the stage scores. Strict propriety of the rank stage is Theorem 1 applied to densities on $[0,1]^d$, the transform $x \mapsto u$ being fixed once the margin reports are given. The final clause is Sklar's theorem applied to the true joint law. ■

This is the two-stage estimation logic of inference functions for margins (Joe 1997), and the factorization underlies out-of-sample copula comparison (Diks et al. 2010); here it is read as a market design. The rank stage is the multivariate Residual operator: under correct margins the rank vector has uniform margins and its joint law is the dependence structure alone, so a rank-settled market elicits dependence separately from level. When the margins are correct the settled object is invariant to monotone transformations of the coordinates; the settlement transform itself moves with the reported margin maps. The microprediction platform ran exactly this factoring: alongside univariate z-streams it operated bivariate and trivariate streams in which community-implied percentiles were embedded by a space-filling curve into one settled scalar (Cotton 2022), a copula market in production. Sample messages compose with the factoring: rank each coordinate of a submitted cloud through the margin CDFs and the result is a cloud on $[0,1]^d$ for the copula stage, with the §6 seam discipline applied at that stage's settlement.

8 Implementation notes

The mechanisms of the catalogue are implemented against a single interface patterned on the skaters time-series contract (Cotton 2026b), a successor to timemachines (Cotton 2021): every stage consumes and emits the same distributional type and threads its own state, so the signature of §4 is the forecasting contract with wealth in place of model state. The closure of the message type is what makes a small operator set sufficient; the transforms, ensembles, and residual constructions of §4 all consume and emit the one type. Scores are implemented in loss form; the text uses reward form.

9 Open problems

1. *Conservation of edge.* Wealth conservation is automatic, a sum of stage-level zeros, and is not the invariant. The invariant that can fail is the edge a truthful participant holds over the price, $D_G(q,\pi)$. For the logarithmic score the edge is $\mathrm{KL}(q\|\pi)$ and the data-processing inequality settles it: an interface channel cannot increase it, and preserves it exactly when the channel is sufficient for the pair (Csiszár 1967). No such inequality holds for a general Bregman edge. For the quadratic generator, merging the first two of three outcomes sends $\pi = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ and $q = (\frac{13}{30}, \frac{13}{30}, \frac{2}{15})$, with $D_G(q,\pi) = \frac{3}{50}$, to a pair with $D_G(qK, \pi K) = \frac{2}{25}$: coarsening increased the Brier edge. Open: characterize the generators and channels for which $D_G(qK, \pi K) \le D_G(q,\pi)$, and quantify the contraction or amplification of edge through a given interface.

2. *Residual markets.* Proposition 7 gives the single-step correction; a chain of residual markets is stagewise boosting with wealth as the learning rate. Open: the microstructure (who funds each residual pot, and when it settles relative to the stage before), whether a forecaster free to enter several stages prefers withholding information upstream to sell it downstream, and whether the iterated correction converges to the true conditional law as boosting does. Split-conformal prediction is the one-participant degenerate case, with its rent $I(R; X)$ as the value left on the table (Theorem 9). The closest live analogue is in point prices: a virtual bid in a two-settlement electricity market is written on the day-ahead stage's pricing error, and its convergence bidders are the residual stage's informed entrants (Jha and Wolak 2023). The distributional version is the gap.
3. *Copula markets.* A rank-settled stage elicits dependence separately from margins (Proposition 13), and under correct margins its settled object is invariant to monotone transformations of the coordinates. The racetrack's win and exotic pools price margins and joint orders in parallel books on a finite outcome space, with consistency left to arbitrage and the Harville map as the bridge (Harville 1973; Hausch et al. 1981); the rank-settled stage differs by construction, chaining through the reported margins so the dependence market cannot disagree with them. Open: the equilibrium when the same participants trade margin and copula stages, and the choice of embedding for $d \geq 2$, the space-filling curves as deployed against the random one-dimensional projections of the companion paper.

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